

Aditi Kulkarni

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SUMMARY

Senior engineer with 8+ years building high-throughput Python backend systems and data pipelines across API services, NLP/data workflows, and large-scale automation, plus Java/Scala and Golang. Owns services end to end: design, streaming/data infra, observability, and safe rollouts.

TECHNICAL SKILLS

Languages: Python 3, Golang, Java, Scala, SQL

Data & Messaging: Kafka, Spark, ClickHouse, Redis, Elasticsearch, MongoDB, GCP BigQuery, GraphQL

Infra & Tooling: Kubernetes, Docker, AWS, Ansible, CI/CD, Prometheus, Grafana, Git, JIRA, Agile

WORK EXPERIENCE

Walmart

Sunnyvale, CA

Senior Software Engineer

Nov 2024 – Present

- **Driver Language Preference Enablement (Python):** Architected a Python NLP-powered translation pipeline delivering Spanish language support to 30% of Walmart's driver base; built as modular, pytest-covered services and mentored developers across teams for parallel delivery
- **Real-Time Events Tracking & Incident Detection:** Built a PySpark/Scala streaming pipeline processing >3M Kubernetes events/day into ClickHouse and Redis behind a GraphQL API (<3 ms latency); cut incident mitigation to <5 minutes via Prometheus/Grafana
- **Local Rate Limiting:** Designed and rolled out per-service rate limiting across the microservice fleet; analyzed traffic patterns to set thresholds and built dashboards, preventing 3 major production incidents from resource exhaustion
- **Shopper Fraud Reduction – Last Mile Delivery:** Built guardrails for shopper drop workflows (validation checks, pharmacy-order edge cases) behind feature flags, achieving a 95% reduction in orders stuck at pick-complete

Software Development Engineer III

Jun 2022 – Nov 2024

- **Distributed Deployment Automation (Python):** Built Python automation deploying plugins to 15,000+ applications via multi-pod Kubernetes parallelism; cut a 3-day, 4-engineer process to under 5 hours with a self-serve Slack bot
- **Dependency Mapping & Remediation System:** Built an end-to-end Java/Golang service across 60,000+ applications, exposing dependencies via GraphQL over MongoDB and Elasticsearch; reduced MTTD/MTTR from >6 hours to under 1 hour

Red Hat Inc

Los Angeles, CA

Software Engineer

Jul 2019 – Jun 2022

- **Business Logic Automation – Home Mortgages Platform:** Led development of a rules-driven application embedding 1,500+ business rules with automated JUnit tests, cutting loan processing from 1 week to under 1 hour and reducing manual intervention to critical errors only

KPMG India

Data Analyst

Jun 2016 – Jul 2017

- Built customer personas and journey maps through exploratory data analysis in Python, identifying service gaps and driving a 40% improvement in operational efficiency

PROJECTS

- **TF-MNIST** (*Python, TensorFlow, Docker, S2I, Tekton*) — Convolutional neural network classifying MNIST digits with TensorFlow (tf.data/tf.estimator); containerized via Source-to-Image and deployed through Tekton CI/CD pipelines
- **YouTube Trending Videos NLP** (*Python, pandas, NLP, Jupyter*) — NLP analysis of the YouTube trending dataset, extracting title, tag, and metadata features that predict what makes a video popular

EDUCATION

University of Southern California

Los Angeles, CA

M.S. in Electrical and Computer Engineering

2017 – 2019

Birla Institute of Technology and Science (BITS)

Pilani, India

B.E. (Hons.) Electrical and Electronics

2012 – 2016

PUBLICATIONS

Kulkarni et al., “Robotic Assistance in Indoor Navigation for People Who are Blind,” Carnegie Mellon University Robotics Institute, funded by NSF National Robotics Initiative (IIS-1317989) ([Link](#))